

Tech & Teach

Digital Chip Design Program

Design RTL

names:

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Task :04

2026\07\06

Task topic:

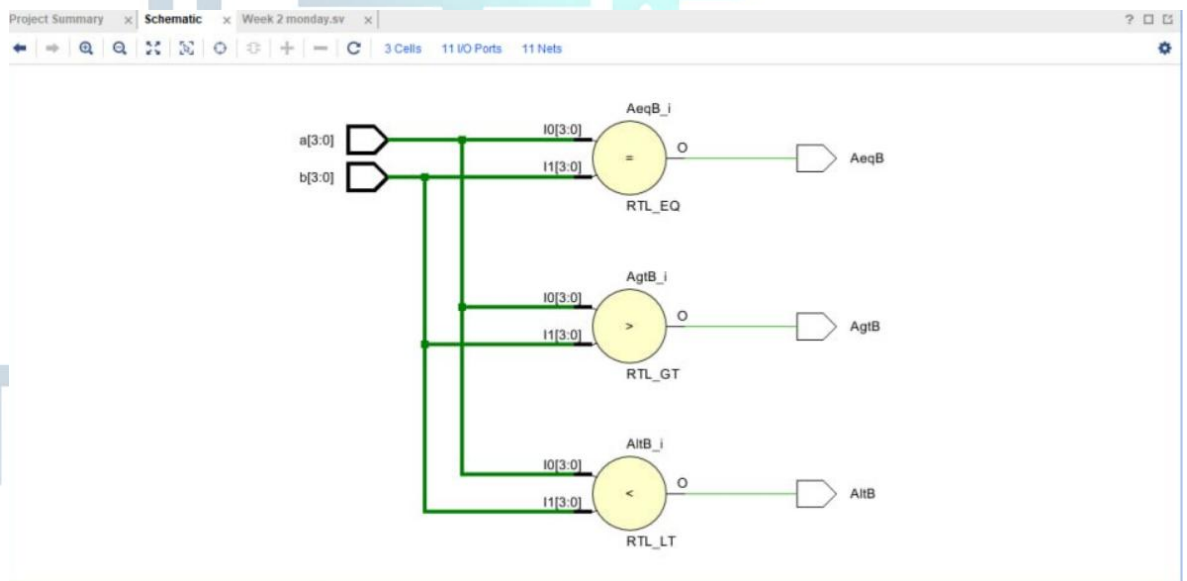
Module instantiation

Version 1.0

Task: 1.1

```
Project Summary x Schematic x Week 2 monday sv x
C:\Users\HUAWEI\Desktop\Wvado\Week 2 monday\Week 2 monday.srcs\srcs_1new\Week 2 monday sv

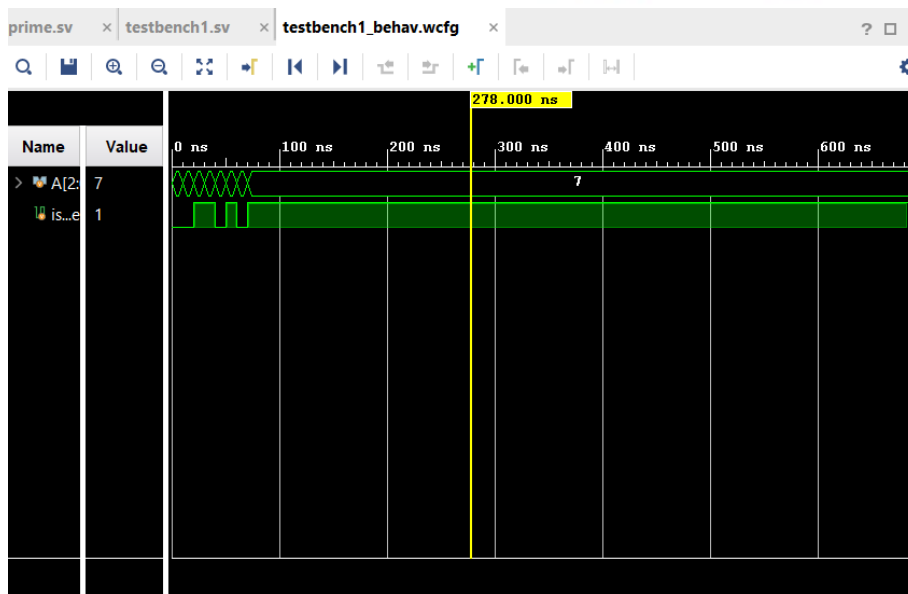
1 module comparator(AeqB, AltB, AgtB, a, b):
2   input [3:0] a,b;
3   output AeqB, AltB, AgtB;
4
5   assign AeqB = (a == b) ? 1 : 0;
6   assign AltB = (a < b) ? 1 : 0;
7   assign AgtB = (a > b) ? 1 : 0;
8 endmodule
```



Task: 1.2

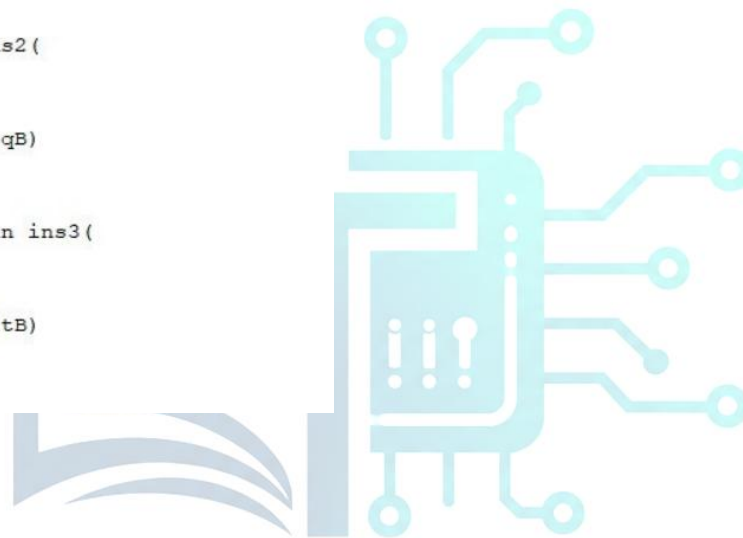
```
prime.vv x testbench1.vv x testbench1_behav.wcfg x
C:/Users/HUAWEI/project_6/project_6/srcs/sources_1/new/prime.vv
1 module prime(
2
3     input logic [2:0] A,
4     output logic isPrime
5
6 );
7
8 assign isPrime = (A[2]) ? A[0] : A[1];
9
10 endmodule
```

```
1 timescale 1ns / 1ps
2
3 module testbench1();
4     logic [2:0] A;
5     logic isPrime;
6     prime dut(
7         .A(A),
8         .isPrime(isPrime)
9     );
10    initial begin
11        A = 3'b000;
12        #10 A = 3'b001;
13        #10 A = 3'b010;
14        #10 A = 3'b011;
15        #10 A = 3'b100;
16        #10 A = 3'b101;
17        #10 A = 3'b110;
18        #10 A = 3'b111;
19    end
20 end
```

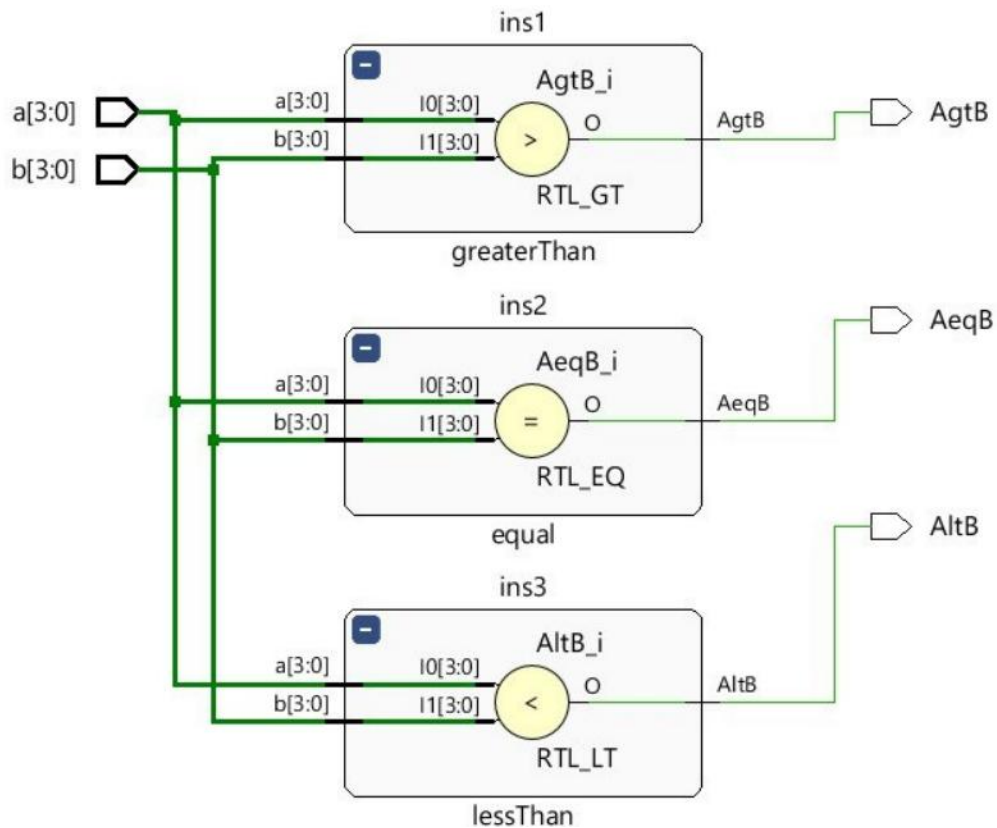
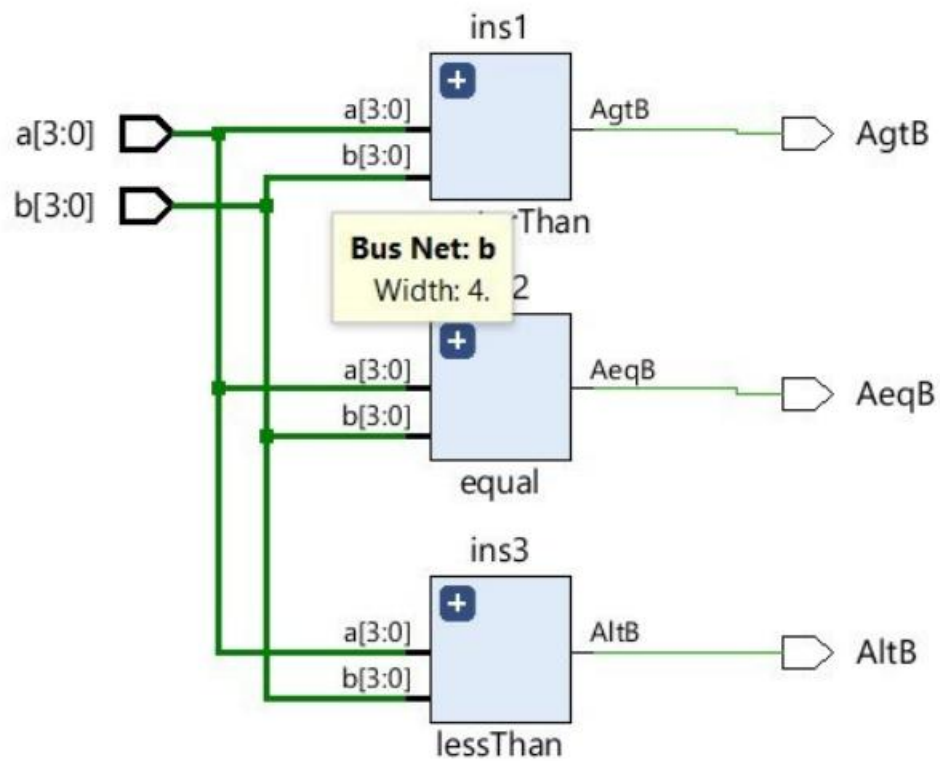


Task: 1.3

```
1
2 module Top(
3   input logic [3:0]a,b,
4   output logic AgtB , AeqB , AltB
5   );
6
7   greaterThan ins1(
8     .a(a),
9     .b(b),
10    .AgtB(AgtB)
11  );
12
13  equal ins2(
14    .a(a),
15    .b(b),
16    .AeqB(AeqB)
17  );
18
19  lessThan ins3(
20    .a(a),
21    .b(b),
22    .AltB(AltB)
23  );
24 endmodule
--
```



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```

1
2 `timescale 1ns / 1ps
3 module Top_tb();
4     logic [3:0] a,b ;
5     logic  AgtB , AeqB , AltB;
6
7     Top dut(
8         .a(a),
9         .b(b),
10        .AgtB(AgtB),
11        .AeqB(AeqB),
12        .AltB(AltB)
13    );
14
15    initial begin
16        a = 1; b = 0 ;
17        #40 a = 1; b = 1 ;
18        #40 a = 0; b = 1 ;
19    end
20
21 endmodule
22

```

code of testbenche



the simulation

Work distribution

Section	shaher	Suliman	Abdullah
Report	✓		
Task 1.1			✓
Task 1.2	✓		
Task 1.3		✓	

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